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Bright Learn Tutorials

Data Analytics

Exercise 1: SQL Fundamentals

Database: employees_db

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions & Answers:

1. SELECT Statement

Write a SQL query to retrieve all columns from the employees table.

Answer:

```
SELECT *

FROM employees_db;
```

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston

8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

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2. SELECT DISTINCT Statement

Write a SQL query to find all the unique departments in the employees table.

Answer:

```
SELECT DISTINCT department

FROM employees_db;
```

departments
IT
HR
Finance
Marketing

---

3. ORDER BY Statement

Write a SQL query to retrieve all employees' first and last names, ordered by salary in descending order.

Answer:

```
SELECT first_name, last_name, salary

FROM employees_db

ORDER BY salary DESC;
```

first_name	last_name
Emily	Davis
Mike	Johnson
Robert	Wilson
John	Doe
Sarah	Brown
Daniel	Clark
David	White
Jessica	Moore
Laura	Hall
Jane	Smith

---

4. LIMIT Statement

Write a SQL query to retrieve the top 5 highest-paid employees.

Answer:

```
SELECT *

FROM employees_db

ORDER BY salary DESC

LIMIT 5;
```

id	first_name	last_name	department	salary
6	Emily	Davis	IT	62 000
3	Mike	Johnson	Finance	60 000
7	Robert	Wilson	Finance	59 000
1	John	Doe	IT	55 000
4	Sarah	Brown	IT	53 000

---

5. WHERE Statement

Write a SQL query to find employees who work in the IT department.

Answer:

```
SELECT *

FROM employees_db

WHERE department = 'IT';
```

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55 000	2018-06-15	New York
4	Sarah	Brown	IT	53 000	2021-03-25	New York
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Fransico

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6. AND Statement

Write a SQL query to find employees who work in the Finance department **AND** have a salary greater than 58,000.

Answer:

```
SELECT first_name AS Name, last_name AS Surname, department, salary

FROM employees_db

WHERE department = 'Finance'
```

AND salary > 58000;

Name	Surname	department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000

---

7. OR Statement

Write a SQL query to find employees who work in the HR department **OR** the Marketing department.

Answer:

```
SELECT first_name, last_name, department

FROM employees_db

WHERE department = 'HR'

OR department = 'Marketing';
```

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

---

8. NOT Statement

Write a SQL query to find employees who **do not** work in the IT department.

Answer:

```
SELECT first_name, last_name, department

FROM employees_db

WHERE NOT department = 'IT';
```

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

---

9. IN Statement

Write a SQL query to find employees who are in the HR, IT, or Finance departments.

Answer:

```
SELECT first_name, last_name, department

FROM employees_db

WHERE department IN ('HR', 'IT', 'Finance');
```

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

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10. Combining Conditions

Write a SQL query to find employees who are in the IT department, have a salary greater than 50,000, **and** are located in New York.

Answer:

```
SELECT *

FROM employees_db

WHERE department = 'IT'

AND salary > 50000

AND city = 'New York';
```

id	first_name	last_name	department	salary	hire_date	City
1	John	Doe	IT	55 000	2018-06-15	New York
4	Sarah	Brown	IT	53 000	2021-03-25	New York

---

11. Combining WHERE, AND, and ORDER BY

Write a SQL query to retrieve the first and last names of employees who work in the **Finance** or **Marketing** department, **earn more than 52,000**, and **order the results by salary in descending order**.

Answer:

```
SELECT first_name, last_name, department, salary

FROM employees_db
```

```
WHERE (department = 'Finance' OR department = 'Marketing')

AND salary > 52000

ORDER BY salary DESC;
```

first_name	last_name	department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53 000

---

12. Combining SELECT DISTINCT, WHERE, and IN

Write a SQL query to find all the **unique cities** where employees work, **excluding those in the IT and HR departments**.

Answer:

```
SELECT DISTINCT city

FROM employees_db

WHERE department NOT IN ('IT', 'HR');
```

city
Los Angeles
San Francisco
Houston
Chicago

---

13. Combining WHERE, NOT, AND, and ORDER BY

Write a SQL query to retrieve employees who **are NOT in the Finance department**, **have a salary greater than 50,000**, and **order the results by hire date in ascending order**.

Answer:

```
SELECT *

FROM employees_db

WHERE department <> 'Finance'

AND salary > 50000

ORDER BY hire_date ASC;
```

id	first_name	last_name	department	salary	hire_date	City

5	David	White	Marketing	52 000	2016-04-10	San Fransico
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
1	John	Doe	IT	55 000	2018-06-18	New York
4	Sarah	Brown	IT	53 000	2021-03-25	New York
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago

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14. Combining WHERE, OR, IN, and LIMIT

Write a SQL query to find the **first 3 employees** who work in either **Chicago or Los Angeles** and **belong to the IT or Marketing department**.

Answer:

```
SELECT *

FROM employees_db

WHERE (city = 'Chicago' OR city = 'Los Angeles')

AND department IN ('IT', 'Marketing')

LIMIT 3;
```

id	first_name	last_name	department	salary	hire_date	City
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
NULL	NULL	NULL	NULL	NULL	NULL	NULL